AI & ML.

* Artificial Intelligence :-

→ Branch of CS in which such machines are developed that can perform a wide range of tasks exactly and do not require human intelligence.

→ In 1956, "John mccarthy" first addpted the term AI. Therefore, father of AI.

→ Humanoid = Human like not human

→ Developed to make these humanoids Robots

→ In 1958, John mccarthy developed the first AI language. LISP (List programming).

→ Other languages: Prolog (programming in logic), Java, Python, R language, Julia, SmallTalk, Haskell, C++, Scala etc.

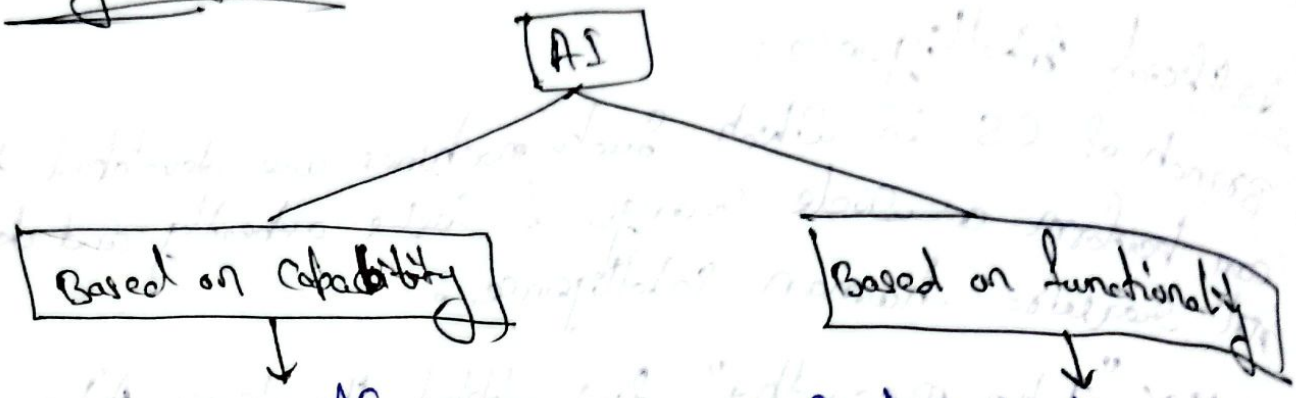
→ AI term includes :-
- learning
- Reasoning.
- Problem solving

* AI branches :-

(a) machine learning (Deep learning)
(b) Expert system
(c) fuzzy logic.
(d) Neural logic.
(e) Natural language
(f) Robotics.

first AI programme = The logic therapist (1955)
By = Allen Newell.

* Type of AI :-



* Based on Capability :-

(1) Narrow AI
→ Do specific task.

ex:- (a) Virtual / Digital Assistant - Alexa, Siri, Cortana, Bixby, Google Assistance.
(b) Face Recognition / Image Recognition System.
(c) Tesla Self Driving Car.
(d) Email Spam Filter.

(2) General / Strong AI :-
→ Can do exactly work like Human beings.
→ Still in development phase.

(3) Super AI :-
→ more Strong than Human beings.
→ Still in Imaginary phase.

* Based on Functionality :-

(1) Reacting machine :-

→ focus on present scenario, ignore past experience & memory

eg:- IBM chess game player Deep Blue (1997)

Defeated Rank 1 chess player :- Garry Kasparov.

(2) Limited memory :-

→ Donot ignore past exp. and memory

eg: Self driving car.

(3) Theory of mind :-

→ AI will work as Human being.

(4) Self Awareness :-

→ Stronger machines than human beings.

* Evolution of AI :-

1943 → first AI program was written unknowingly in mark-1 computer.

↓

1950 → Alan Turing launch Book/Article : "Computer machinery & intelligence"

↓

1956 → John Mccarthy workshop :- Study of AI.

↓

1958 → LISP (list programming)

↓

first AI lang.
Developed by → John Mccarthy
Introduced term "Humanoid".

1958 → lick linder define & Human machine relationship

↓

1962 → Start first Robotic Company "UNIMATION".
by - Joseph Engelberger
→ father of Robotics.

1966 → ELIZA = First AI Program (Chat bot)
by - Joseph Engelberger.

1973 → WABOT-1 = first Humanoid Robot
Japan Waseda University.

1977 → IBM Deep blue = Chess Game player
Defeated world's Ranks player = Gary Kasparov.
first Intelligent Humanoid.

14 Feb 2016 → Sophia AI Robot Invented.
→ face Recognition
→ Hongkong Company "Hanson Robotics".
→ first Robot to get Citizenship.
→ 2017 UAE Citizenship.

* Agent :-
→ An independent entity who observe the environment to achieve goal.

Agent = Architecture + Program.

5 types :-

(a) Simple Reflex Agent.
(b) Model Based Reflex Agent.
(c) Goal Based Agent.
(d) Utility Based Agent.
(e) Learning Agent.

(a) Simple Reflexive Agent :-

→ Observe the Environment, take action accordig to condition.
→ Pre programmed.

eg. if temp > 28
AC on
else
Ac off.

(b) Model Based Reflexive Agent :-

→ Observe environment partially. Take action.

eg: Self driving cars.

(c) Goal Based Agent :-

→ Extended version of model based reflexive Agent.
→ Searching & planning.
→ eg: Google: Waymo Self driving car (2009)

(d) Utility Based Agent :-

→ Happy and unhappy state based Agent.
→ choose optimized both from multiple both. (eg:- Price comparision)

(e) Learning Agent :-

→ learn from past experience
→ Human beigne.

* Domains of AI :-

(A) Data Science.
(B) Computer Vision.
(C) NLP (Natural language Processing).

(A) Data Science:-

→ Study of data.
→ Can be audio, video or textual.
→ Data System & processing principles are applied.
→ mathematical and statical

(B) Computer Vision :-

→ Image and objects are identified.
eg: face recognition apps (Google lens, google translate).

(C) NLP :-

→ used for communication between computers & human being.

* Searching Algorithm :-

Informed	Uninformed
1) Heuristic Searching Algo.	Blind/Brute force searching Algo.
2) knowledge based algo	Doesn't use knowledge.
3) S/n is found Quickly	Solution is found late.
4) No guarantee of completion	Guarantee of completion.
5) Direction for S/n are known	unknown.
6) A*, Graph search	BFS & uniform cost search also, DFS.

Machine learning:-

* It is a subsystem of AI.
→ ML learns by itself, so there is no need to write separate algorithm programs.
→ 3 types: (A) Supervised learning.
(B) Unsupervised learning.
(C) Reinforcement learning.

(A) Supervised learning:-

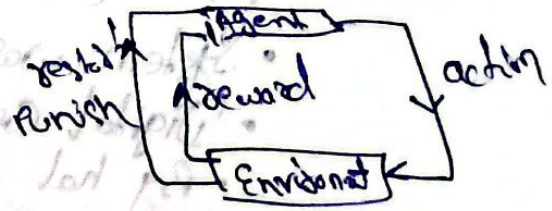
→ In this type of learning, the input data is given along with labels, which contain details about the data.
→ There are 2 types.
(a) Regression → relationship b/w 2 or more variable.
(b) classification → works on Yes/No.

(B) Unsupervised learning:-

→ Unlabelled data is used.
→ Data is trained by itself.
→ Hidden patterns are also found.
→ Algorithm: K-means cluster Algorithm, Hierarchical cluster Algo.

(C) Reinforcement learning:-

→ Based on reward and punishment mechanism.
→ Trained by itself.
→ Feedback-based.
→ Hit & trial.
→ Algorithm:- value-based, Policy-based.



** Block chain :-

(A → It is a type of ledger that keeps a record of transactions happening in a peer-to-peer network.

(→ Any information transferred through blockchain is encrypted / secured, so it cannot be changed.

(A) → In a block, there is a data structure for digital transaction. Data / transaction are stored with a hash value so that no changes can be made to them.

→ When multiple blocks are connected together, a secured chain is formed, which is called a blockchain.

(B → Data, records, contracts and documents can be stored.

→ There is no privacy breach.

→ No centralized control.

(C → No specific owner.

* History of Blockchain :-

- 1991 :- Stuart Haber & Scott Stornetta proposed the concept of cryptographically secured block (Secured blockchain).
- 1992 :- The first secured blockchain was created using Merkle tree. Ralph merkle (1979)
- 2004 :- RPoW (Reusable proof of work)
  - System was created to store digital transactions.
  - Digital cash was stored.
  - By hal finney.
- 2008 :- Satoshi Nakamoto,
  - Started Crypto currency.

* Content of Block :-

(a) Data = Record / Transaction
(b) Timestamp = Creation time of block.
(c) Hash value = Block secure.
(d) Previous hash = Hash value of the prev. block.

* Fuzzy logic :-

→ This is a method of reasoning, which is similar to human reasoning.
→ Fuzzy → Not clear.
→ first introduced by Lotfi Zadeh.
→ Any general computer only makes decision in yes/No. but fuzzy logic also includes possibilities b/w yes and No.

Ex Is Raju a teacher → Yes → fuzzy (nearly for. yes, sometimes/no)
→ No

* Advantages :-

→ work with any type of input.
→ less memory needed.
→ Its structure is easy to understand.
→ Provides efficient solutions to complex problems.
→ users can easily add or delete rules.

* Disadvantages :

→ The soln is not always correct.
→ Sometimes it very confusedly.
→ There is no systematic approach to its design.
→ It is very slow, so output takes time.

* Applications:-

→ Controlling the height (altitude) of spacecrafts and satellites.
→ Decision making support systems.
→ Assuming target in data.
→ Cooking strategies in microwave.
→ Traffic control.

* Generative AI :-

→ This is a technology that can process existing data and create new things.
→ Based on machine learning & Deep learning.
→ It learns the pattern in data and generates new data with similar patterns
ex:- chat gpt, DALL-E.
→ Neural Networks are used, which work like the human mind.